

Mini-Game Design Guide

Level 1

Complete both level 1 & level 2 before you begin coding.

This guide walks you through four design stages for each level.

For each stage, answer the guided questions in writing before you write any code.

Your answers will help you shape the software architecture.

Expected time: approximately 45–60 minutes.

The completed template should be submitted alongside the project report.



Stage 1: Interaction

Create interaction: **What action triggers the gameplay cycle?**

Design Answers

What is the primary action the player performs in the game?

The player moves the character to collect coins and avoid traps

What task is the player aiming to achieve through this action?

To collect all coins and reach the next level without hitting traps

What triggers this action?

Keyboard arrow keys (UP, DOWN, LEFT, RIGHT)

What on-screen element allows performing this action?

The player character controlled on the game screen

Checklist

- ✓ The player's main action can be explained by one clear sentence.
- ✓ There is at least one interactive component added to the interface.
- ✓ I can identify the starting point of the gameplay cycle.

Collect all bananas and avoid traps



Stage 2: Response

Make it responsive: **How does the game respond to the player's action, and what progress results from that response?**

Design Answers

What feedback does the game provide after the action?

The player's position updates, coins disappear when collected, and traps end the game if hit

What resources increase or decrease as the player progresses?

Score increases when collecting coins, and time decreases every second

How is the player's progress in the task presented in the interface?

Through labels showing Coins, Level, and Time

When should progress be recorded in the game?

Progress is recorded immediately after collecting coins or hitting traps

Checklist

- ✓ I can name at least one resource that reflects player progress.
- ✓ The interface updates when progress is made.
- ✓ I can identify when to record progress, which is after important actions.



Stage 3 — Logic

Definethellogic: **What rules cause changes in the game state?**

Design Answers

What is the primary rule of the game?

Collect all bananas and avoid traps

What component maintains the current state of the game?

The Controller class

What information does it retain?

Player, score, level, time, coins, and traps

What rule decides how the game state changes as the task progresses or is completed?

Collecting coins, hitting traps, or time ending changes the game state

Checklist

- ✓ The primary rule of the game is clear.
- ✓ The component that maintains the current state is identified.
- ✓ I can list the main information it retains.
- ✓ I can explain what causes the state to change.



Stage 4 — Challenge

Shapethechallenge: **How does the game get harder during gameplay, and encourage stronger strategy?**

Design Answers

What is the most challenging moment in the game?

Level 2 is the hardest part

How does the game change during gameplay?

The difficulty increases in Level 2

How do those changes make the task harder and affect the player's action?

The player must move faster and more carefully

What progress or state should be restored when the game starts again?

Score, level, and collected coins should be restored

Checklist

- ✓ I can clarify what makes the game more challenging during gameplay.
- ✓ The player needs to adapt strategy in order to succeed.
- ✓ The game is designed to be played several times.
- ✓ I can name what progress or state should be restored when the game starts again.

Mini-Game Progression Guide

Level 2

Use this section to build and expand Level 2 based on Level 1.

Do not repeat your answers from Level 1.



Interaction: How does the player's action change in Level 2?

Design Answers

What are the changes in the player's action or task in Level 2?

The player must avoid more traps and collect coins faster

Do these changes reflect a more complex, faster, or different task?

Yes, Level 2 is faster and more challenging

Checklist

✓ The action in Level 2 is noticeably different from Level 1.



Response: How does the response of the game change in Level 2?

Design Answers

What are the changes in the feedback or resource rules that affect task progress in Level 2?

Level 2 had reduced lighting (darkness), making it harder to play

How are these changes presented in the interface?

This is shown through the level, score, timer, and darker visibility

Checklist

✓ The response in Level 2 reflects increased challenge.



Logic: What game rules are changed in Level 2?

Design Answers

What new rule is included in Level 2 and how does it affect the task progress or completion?

Level 2 adds darkness (limited vision), making coins harder to collect and increasing trap risk

How does this new rule impact the game state?

A dark overlay with limited vision around the player changes movement and navigation

Checklist

✓ Level 2 adds at least one rule.

--	--

 Challenge: How does Level 2 develop into a more challenging level than Level 1?	
Design Answers What causes the task in Level 2 to be harder? <u>Reduced visibility makes traps and coins harder to see, increasing difficulty</u> How does the player need to change strategy to succeed? <u>The player must move slowly and carefully instead of rushing</u>	Checklist ☒ Level 2 requires a different strategy than Level 1.

Progress & Continuity: What transfers from Level 1 to Level 2?	
Design Answers What task progress, resources, or game state goes on from Level 1 to Level 2? <u>Score, coins, traps, and gameplay continue from Level 1</u>	Checklist ☒ There is a clear connection between Level 1 and Level 2 progress.

Final Reflection	
Design Answers In one sentence, describe how Level 2 expands on Level 1. <u>Level 2 increases difficulty by adding darkness while keeping the same objectives</u>	